

The College Laboratory: Wing-T, Wishbone, Air Raid, Spread Option, and the RPO

Why so much of Sunday's playbook was invented on Saturday, and what the NFL kept

Saturday night, third-and-6. The offense snaps the ball eleven seconds after the last play ended. Four receivers stand spread across a field whose hash marks seem to sit almost on the sideline; the ball is on the left one, and the whole offense seems to have drifted to the wide side of the field. The quarterback, five yards behind the center, puts the ball in his running back's belly, watches a defensive end, pulls the ball, takes two steps, and flips it to a slot receiver bowing backward toward the sideline. Twelve yards. The next afternoon you watch an NFL team run almost the same play: same read, same flip, same receiver. The NFL version is a little quicker, the line looks a little different, and the announcer calls it "a college concept." He is right, and he could have said the same thing about half the plays in the game. Where did they come from, why does college football keep inventing them, and why do some make the jump to Sunday while others never do?

You'll be able to:

- Explain the four forces that make college football the sport's laboratory: talent gaps across hundreds of programs, wide hash marks, rules that tolerate the run-pass blur, and roster churn and tempo.
- Trace the main offensive lineages (split-T and veer, wishbone and flexbone, Wing-T, BYU's passing game, the Air Raid, the spread offense, the spread option, the pistol, tempo and the RPO) and point to their DNA in an NFL play.
- Trace the defensive answers (the 3-3-5 stack, pattern-matching quarters, tite and mint fronts, and the three-high safety defense) and say which problem each was built to solve.
- Say when and how each idea crossed into the NFL, and why some never did.
- Read a college broadcast with the rulebook differences in mind (hash width, one foot in bounds, three yards downfield, a 15-yard pass-interference cap, the college clock, overtime, targeting and more), and predict how each one changes what the teams do.

You'll need

- **1978 and the Passing Revolution:** the NFL side of the story up to 2000, from the Mel Blount rule and Air Coryell to the West Coast offense and the run-and-shoot. This chapter tells the college side of the same decades and carries it forward.
- **Option Football:** the read key, give and keep, the zone read, the inverted veer, the triple option's dive, keep and pitch, and the count (#1 dive key, #2 pitch key, #3 alley, #4 deep). This chapter traces where those plays came from rather than re-teaching them.
- **Run-Pass Options:** the conflict defender, pre- and post-snap RPOs, the bubble RPO, and the one-yard ineligible-downfield rule that makes NFL RPOs look different from college ones.
- **Quarters and Match Coverage:** two deep safeties who read the receivers and fit the run from depth; the read safety; pattern matching; Cover 6.

How to read the diagrams. Offense is blue circles, defense orange squares; *left* and *right* mean the offense's left and right, and the offense always moves up the page. Q is the quarterback and R the running back; X and Z are wide receivers on the line, H and F slot receivers a step off it, and Y a tight end. In the option backfields, F (or B) is the fullback and H or A a halfback or A-back. The linemen are blank circles. On defense, numbers are technique alignments (a 5 sits on the tackle's outside shoulder, a 3 on the guard's), N is a nose tackle and E an end; S, M and W are the Sam, Mike and Will linebackers; CB is a cornerback, NB a nickel back, SP and BA the 3-3-5's two hybrid safeties (the *spur* and *bandit*), and SS and FS safeties. Lines ending in a short bar are blocks, thick dark arrows are ball carriers, dashed arrows are a play's possible *branches*, a dashed brown arc is a pitch or a throw, and a dashed green line from the quarterback shows whom he is reading. Orange rings mark defenders to watch.

Why college invents and the NFL borrows

Watch enough football and a pattern appears. Almost every offensive idea that changed the NFL in the last thirty years, the shotgun spread, the zone read, the pistol, the RPO, the no-huddle as a way of life, the Air Raid's handful of pass concepts, arrived from college, often a decade or more after it was born there. The traffic runs the other way much less often. That isn't because

college coaches are smarter. It is because college football is a different *environment*, and four features of that environment reward invention.

1. More teams, and bigger talent gaps

The NFL has 32 teams drawing from one talent pool, with a draft and a salary cap built to keep them roughly equal (1978 and the Passing Revolution tells how the cap arrived). College football has more than 130 teams in its top division alone, the Football Bowl Subdivision (FBS), and hundreds more below it, and no mechanism at all to keep them equal.¹ Every year a few programs sign most of the best high-school players, and everyone else has to play them.

That gap is the mother of invention. A coach who can't recruit the bigger, faster players has two choices: lose, or change the problem. Almost every lineage in this chapter starts with a coach in the second group: Bill Yeoman at Houston with an offensive line that couldn't block, LaVell Edwards at BYU with no chance of out-recruiting the Southwest's powers, Hal Mumme and Mike Leach at tiny Iowa Wesleyan, Rich Rodriguez at Glenville State, a Division II school in West Virginia. Their inventions all do the same thing in different ways: they **change the arithmetic** so that the defense's better players have more to defend than they have bodies, or have to make a decision they aren't trained for.

The NFL has talent gaps too, but small ones. When the other team's players are as good as yours, a gimmick that exploits a mismatch is worth less, and a scheme that gets your quarterback hit is worth much less.

2. A wider field

College hash marks are 40 feet apart, 60 feet in from each sideline; NFL hash marks are 18 feet 6 inches apart, the width of the goal posts.² The field is the same size. What changes is where the ball can start, and that changes everything about spacing. With the ball on a college hash, the short (boundary) side is 20 yards wide and the wide (field) side 33. The Game in Fifteen Minutes drew the same four-receiver formation on both fields.

A wide field is a gift to a **horizontal** offense, one that stretches a defense from sideline to sideline. With 33 yards to the field side, a defense that wants to cover four receivers spread across it has to send defenders a long way from the ball, out of the **box** (the area near the ball where runs are defended). With 20 yards to the boundary, the offense can put a single receiver into the short side and dare a cornerback to cover him alone, with no room to help. Option and spread offenses were made for that geometry. The NFL's narrow hashes put the ball near the middle almost every snap, which makes the field more symmetrical and takes away some of the leverage.

3. Rules that tolerate the run-pass blur

College rules have been friendlier to offenses that mix run and pass on the same snap. The biggest difference is the **ineligible downfield** rule from Run-Pass Options: in college an offensive lineman may be up to **three yards** past the line when a pass that crosses it is thrown, and the rule doesn't apply at all to passes caught behind the line; in the NFL, the limit is **one yard** on any forward pass.³ Three yards is room to run-block for a beat and still throw. It is the reason college RPOs can be "run first, throw later," and the reason the college game grew so many plays that are a run and a pass at once. Other college rules nudge the same way: one foot in bounds makes a sideline catch easier, and a defensive pass interference foul costs at most 15 yards, which changes the math of a deep throw (the rules section near the end puts them all side by side).

4. Roster churn, practice limits and tempo

A college roster turns over every four or five years, and the best players leave sooner. NCAA rules cap the hours a team can make players practice in season.⁴ A system that has to be taught to a new quarterback every two or three years, in limited hours, rewards **simplicity**: a small number of plays repeated until they are automatic, which is exactly what the Air Raid and the spread option were. Simplicity, in turn, makes **tempo** possible. A team that runs twelve concepts can call them with one word or one hand signal

¹The FBS had more than 130 member programs in 2025, with the Football Championship Subdivision (FCS), Division II, Division III and NAIA below it: "NCAA Division I Football Bowl Subdivision," Wikipedia: https://en.wikipedia.org/wiki/NCAA_Division_I_Football_Bowl_Subdivision.

²NFL Football Operations, rulebook field diagram ("The location of the inbounds hash marks is 70' 9" for professional football, 60' for college football"): <https://operations.nfl.com/the-rules/nfl-rulebook/>. College hashes have been 40 feet apart since 1993: "Hash mark (sports)," Wikipedia: https://en.wikipedia.org/wiki/Hash_marks. Both are also cited in 01-01.

³NCAA Football Rules (2026), Rule 7-3-10 (no originally ineligible receiver more than three yards beyond the neutral zone until a legal forward pass that crosses the neutral zone is thrown); NFL, *2026 Official Playing Rules*, Rule 7, Section 5 (an ineligible player more than one yard beyond the line before a forward pass is thrown): <https://operations.nfl.com/the-rules/nfl-rulebook/>. 04-06 quotes both rules in full. A 2015 proposal to cut the college limit to one yard was tabled: SI Wire, "NCAA tables proposed ineligible downfield rule" (Mar. 6, 2015): <https://www.si.com/college/2015/03/06/ncaa-ineligible-downfield-rule>.

⁴NCAA Division I Bylaw 17.1.7 limits countable athletically related activities to 20 hours a week (and four a day) in season: NCAA, "Countable Athletically Related Activities" chart: <https://www.ncaa.org/sports/2014/10/6/countable-athletically-related-activities.aspx>.

and snap the ball before the defense has substituted or even lined up. College football's clock (for decades it stopped after every first down) gave fast teams more snaps to use it on.

The filter: what the NFL keeps

Those same four forces explain what *doesn't* cross. The NFL is a filter, and an idea has to survive four tests to get through it.

- **The quarterback test.** NFL quarterbacks are the most expensive players in the sport, and every hit is a risk the team can't insure. Ideas that hit the quarterback on most snaps (the triple option, the wishbone) stop at the border; ideas that hit him occasionally (the zone read) cross as a change-up.
- **The talent test.** NFL defenders are fast enough, and evenly enough spread across the league, that a scheme built on a mismatch has to work against the *best* defense on the schedule, not the average one.
- **The rules test.** The one-yard rule, two feet in bounds and the spot-foul DPI all change how an idea plays on Sunday. Some ideas cross in modified form (the RPO, with quicker throws); some don't.
- **The people test.** Ideas reach the NFL most easily inside players. A quarterback who ran the zone read for four years arrives knowing how; a coach who designed it has to convince an NFL building to change.

The rest of the chapter follows the ideas through that filter. Here is the map.

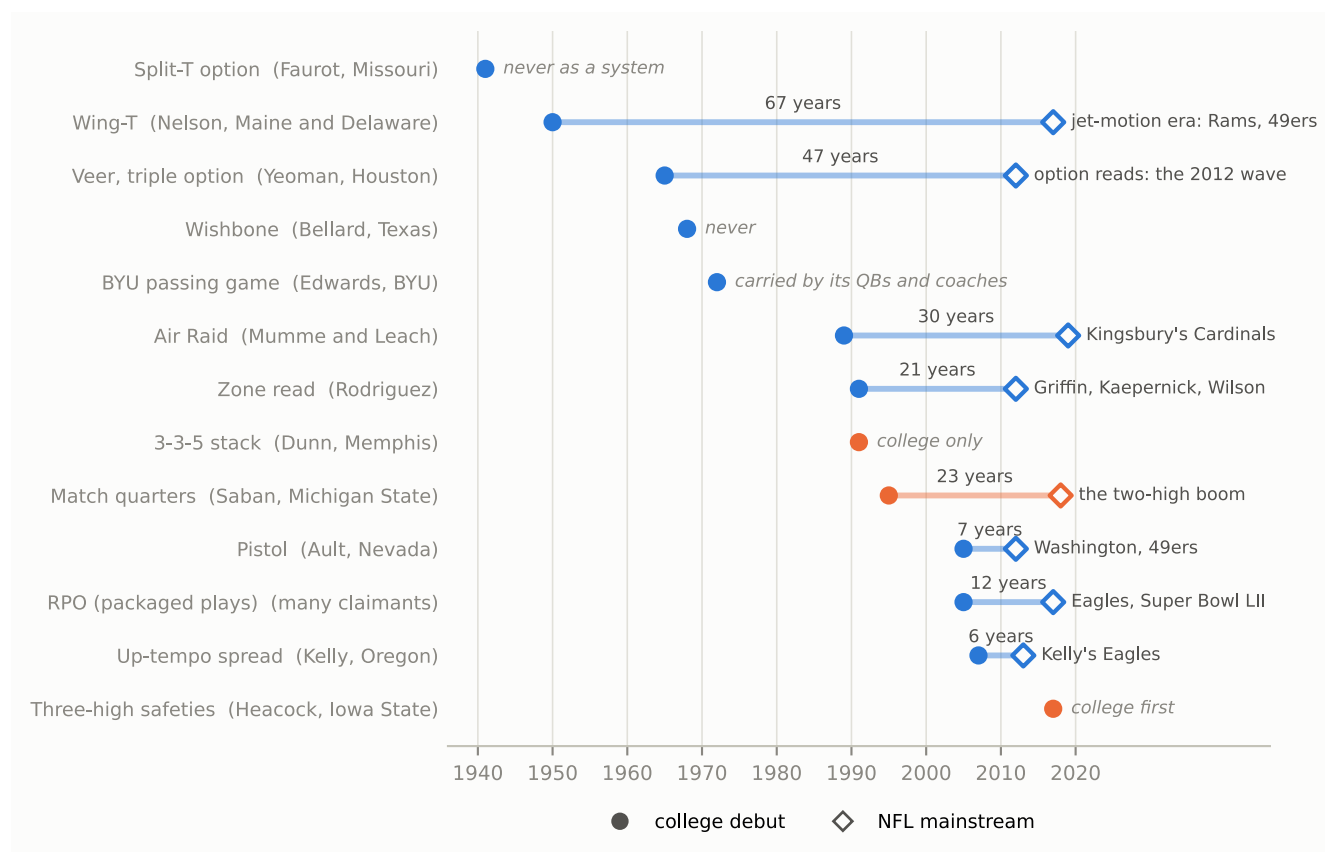


Figure 1: Seventy years of borrowing. Each row is one college invention: the dot is roughly when it debuted in college, the diamond when the idea became mainstream in the NFL, and the label on the line the lag between them. Blue rows are offense, orange rows defense. Rows with no diamond never crossed as a system (the wishbone, the 3-3-5 stack) or crossed only through the people who learned them (BYU's passing game). Dates are approximate by nature: an idea is rarely born on one day, and several (the RPO above all) have more than one claimed inventor; the text and notes give the details.

Two things jump out. First, the lag. Ideas that crossed took a decade or two: the zone read about twenty years, the Air Raid thirty, the Wing-T's motion game closer to seventy. Second, the timing of the crossings is bunched. Almost everything arrived between 2012 and 2019, as the first generation of NFL players who had spent their whole careers in shotgun spread offenses reached the league and as NFL coaches ran out of reasons to resist. We'll come back to why. First, the ideas, in roughly the order they were born.

The option line: split-T, veer, wishbone, flexbone

Option Football taught how the option *works*: leave a defender unblocked, read him, and give the ball to whoever he isn't playing. The history of that idea is a history of college coaches finding new backfields from which to run it.

It begins in **1941** with Don Faurot at Missouri, who widened the T-formation's line splits and built the **split-T**, a series of dives, quarterback keeps and pitches that worked like a modern option. Bud Wilkinson took it to Oklahoma, and Oklahoma's split-T teams won 47 straight games in the 1950s (03-04 has the details). The split-T never became an NFL offense, but it fixed the pattern for the next seventy years: option football is what you run when the other team's linemen are bigger than yours.

The veer (Houston, 1965). Bill Yeoman's Houston teams of the mid-1960s couldn't block the defensive tackles, so Yeoman stopped trying: the play-side tackle released inside and the quarterback read the first defender outside him, giving to a diving halfback or keeping. That is the **veer**, and with the other halfback trailing as a pitch man it is the **triple option**. It made Houston one of the highest-scoring teams in the country: in 1968 the Cougars beat Tulsa 100-6.⁵

The wishbone (Texas, 1968). Two years later, Texas offensive coordinator Emory Bellard moved the split-T's fullback up close behind the quarterback and set the two halfbacks deeper and wider behind him: the **wishbone** of Classic Formations. Darrell Royal's Longhorns won 30 straight games with it; Oklahoma and Alabama copied it; and through the 1970s it was the most feared offense in college football.⁶ Here is what it was built to do.

The wishbone triple option

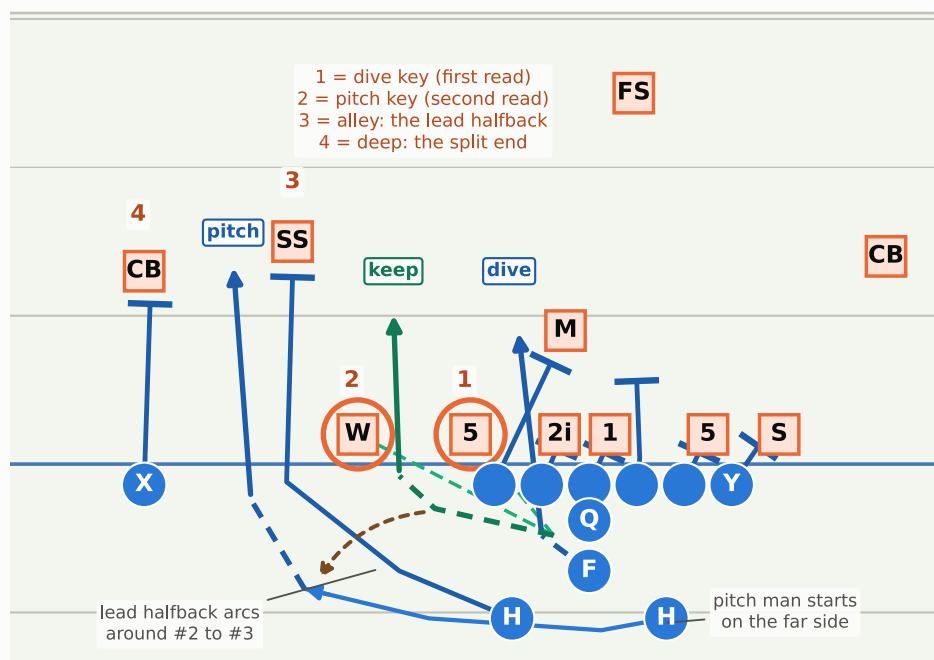


Figure 2: The wishbone triple option to the split-end side (the offense's left): the assignments. The fullback (F) dives at the left guard's outside hip. The left tackle 'veer releases' inside to the Mike linebacker, leaving the 5-technique unblocked as the first read (#1, the dive key). The Will, standing on the line outside him, is left unblocked as the second read (#2, the pitch key). The play-side halfback (H, left) arcs wide around the Will, never touching him, to block the strong safety in the alley (#3), and the split end (X) blocks the cornerback (#4). The other halfback, who starts on the far side of the formation, sprints behind the quarterback to become the pitch man: that is why the wishbone's halfbacks line up deeper than the fullback. The quarterback rides the dive and reads #1 (dashed green), then, if he keeps, runs at #2. Every defender on the play side is blocked or read.

The wishbone's genius was that **every back had a job on every play, to either side**: dive, lead or pitch. The defense had to assign someone to each of three runners on both sides of the formation, with every assignment perfect, and do it without knowing which way the play was going until the snap. In 1971 Oklahoma's wishbone averaged 472.4 rushing yards a game.⁷

⁵"1968 Houston Cougars football team," Wikipedia: https://en.wikipedia.org/wiki/1968_Houston_Cougars_football_team (Houston 100, Tulsa 6, November 23, 1968). On Yeoman and the veer's invention, see the sources in 03-04.

⁶"Wishbone formation," Wikipedia: https://en.wikipedia.org/wiki/Wishbone_formation (Emory Bellard, 1968; Texas's 30 straight wins; Oklahoma and Alabama adopt it); Texas Longhorns lettermen's history, "1968-1970: 30 straight wins and two national championships": <https://www.texaslsn.org/football-dkr-1968-1970>. Also cited in 02-03.

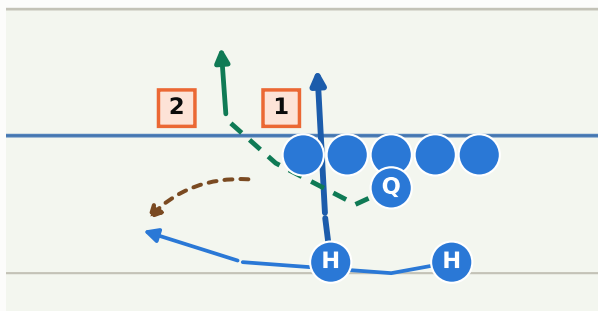
⁷Phi Delta Theta museum, "Jack Mildren" (the 1971 Sooners went 11-1, "with the wishbone averaging a remarkable 472.4 yards rushing per game"): <https://museum.phideltatheta.org/famous/jack-mildren/>; "Jack Mildren," Wikipedia: https://en.wikipedia.org/wiki/Jack_Mildren.

Why did it fade? The same forces that made it, in reverse. **Defenses got faster** through the 1970s and 1980s, and fast defenders who play their assignments shrink the margins the triple option lives on. **The best high-school quarterbacks wanted to pass**, because passers were what the NFL drafted, and the programs that could recruit them stopped needing a talent equalizer. And **it was hard to come from behind with** an offense that threw a dozen times a game. Oklahoma won its last national title with the wishbone in 1985; by the 1990s the formation survived mostly at the service academies.⁸

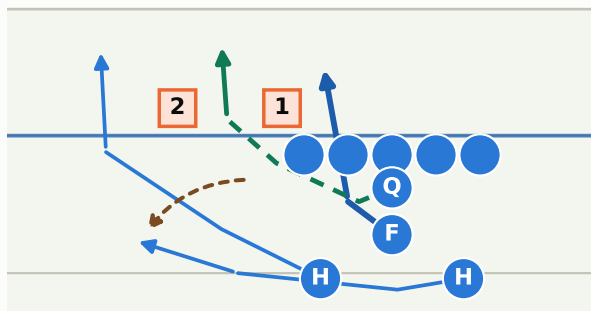
The flexbone (late 1970s and 1980s). The wishbone's last descendant fixed its biggest weakness, width. By standing the two halfbacks just off the line as slotbacks (A-backs), the **flexbone** spread the defense like a passing formation while keeping three runners near the ball; one A-back's pre-snap motion became the pitch. Its invention is disputed (Bellard at Mississippi State, Fisher DeBerry's Air Force, Paul Johnson at Georgia Southern all have claims); Johnson's Navy and Georgia Tech teams are its modern face (02-03).

Put the four option backfields side by side and the idea's survival strategy is plain: the *three threats* never change; only where the threats start does.

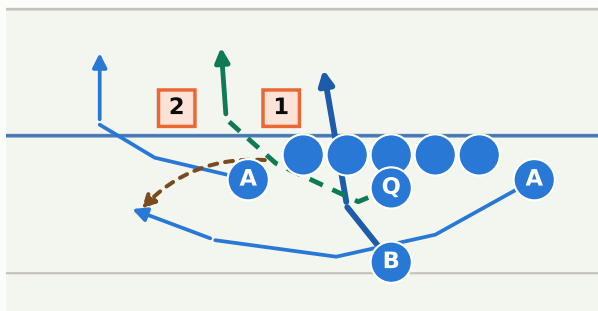
Split-back veer (Houston, 1965)



Wishbone (Texas, 1968)



Flexbone (1980s)



Spread option (2000s)

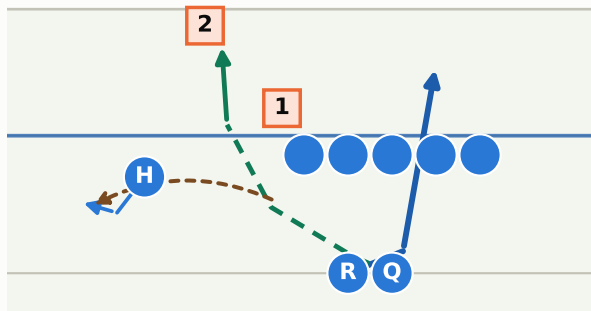


Figure 3: One idea, four backfields: the triple option's three threats to the offense's left in each era (offense only, plus the two read keys, numbered). In every panel the inside runner (thick dark arrow) attacks just inside the dive key (#1), the quarterback (green dashed arrow) keeps outside him, and a third threat (thin arrow and brown arc) works outside the pitch key (#2). Split-back veer: the play-side halfback dives, the backside halfback swings across as the pitch man. Wishbone: the fullback dives, the far halfback pitches, the near halfback leads. Flexbone: the B-back dives, the backside A-back goes in motion to become the pitch man, the near A-back arcs. Spread option: the back runs the inside-zone track the other way (the give), the quarterback keeps, and the third threat is no longer a pitch but a throw: a bubble screen to the slot. The pitch has become a pass.

What crossed into the NFL from this line was never a formation. It was the **idea** in the last panel: a quarterback who reads one defender instead of blocking him, run from the shotgun, with a third threat that can be a throw. That idea needed thirty more years and two more inventions, the spread offense and the zone read, before it could reach Sunday.

⁸Oklahoma's 1985 national champions under Barry Switzer were the last wishbone team to win the title: "Wishbone formation," Wikipedia (as above); "1985 Oklahoma Sooners football team," Wikipedia: https://en.wikipedia.org/wiki/1985_Oklahoma_Sooners_football_team.

i Film room: Oklahoma 1971

The 1971 Sooners, coached by Chuck Fairbanks with Barry Switzer as offensive coordinator, are the wishbone at its peak: quarterback Jack Mildren, halfback Greg Pruitt and the 472-yard rushing average. Their most famous game is the one they lost, 35–31 at Nebraska on Thanksgiving Day, the “Game of the Century” between the top two teams in the country.⁹

What to notice on the replays that circulate online: count the defenders Nebraska puts on each side of the ball, and watch how often a Nebraska defender who did his one job perfectly still ended up tackling a man without the ball. Then watch the pitch man’s spacing on the long runs: four or five yards outside the quarterback and a yard deeper, so the pitch key can never take both.

The Wing-T and BYU: misdirection and the passing equalizer

Not every college lineage was an option. Two others, one running and one passing, fed the NFL more quietly.

The Wing-T (Maine and Delaware, about 1950). Dave Nelson crossed the T-formation with the single wing’s wingback to make the **Wing-T**, and his successor at Delaware, Tubby Raymond, ran it for 36 seasons.¹⁰ Its logic, taught in Perimeter Runs and Misdirection, is **series football**: several plays that start with the same backfield action and end differently, so a defense that flows to the first thing it sees is punished. The buck sweep, the trap, the counter and the waggle all start the same way. The Wing-T itself never became an NFL offense, but its *logic* did, twice: through high-school coaches such as Gus Malzahn, who ran its series from the shotgun at Auburn (below), and through the NFL’s jet-motion era of the late 2010s, when Sean McVay’s Rams and Kyle Shanahan’s 49ers made one backfield picture the start of half a dozen plays (03–05 traces both).

BYU (1972 to 2000). LaVell Edwards took over a Brigham Young program in Provo, Utah, that had no hope of out-recruiting the big programs, and decided to throw. Through the 1970s and 1980s BYU passed when almost nobody else in college football did, spreading the ball to backs and tight ends with timing routes and crossing patterns, and produced a line of record-setting quarterbacks: Gifford Nielsen, Marc Wilson, Jim McMahon, Steve Young, Robbie Bosco and Ty Detmer, who won the Heisman Trophy in 1990. In 1984 BYU went 13–0 and won the national title.¹¹

BYU’s passing game didn’t cross into the NFL as a system. It crossed in **people**, which is the first example of a pattern this chapter keeps finding. McMahon won a Super Bowl with the 1985 Bears and Young two MVP awards with the 49ers; Mike Holmgren, BYU’s quarterbacks coach in the early 1980s, went to San Francisco and later built the Packers’ West Coast offense; Andy Reid played offensive line for Edwards; and Norm Chow, Edwards’s longtime assistant, became the most sought-after offensive coordinator in college football.¹² And BYU’s spacing and crossing routes became the raw material of the next lineage.

! Why it exists

Every lineage so far answers the same problem: **the other team has better players**. The option answers it by reading a defender instead of blocking him (one fewer man to beat), the Wing-T by making the defense guess wrong about where the ball is going (defenders who flow the wrong way don’t count), and BYU’s passing game by spreading the ball to five receivers so a bigger, slower defense has to cover all of them. Each trade is real. The option gives up the passing game and puts the quarterback in harm’s way; the Wing-T gives up width and demands huge practice time for its fakes; the passing game gives up the clock and asks a quarterback to be very good. College coaches could make those trades because their alternative was losing. NFL coaches, whose players are evenly matched, usually take the idea and leave the trade.

⁹“1971 Nebraska vs. Oklahoma football game,” Wikipedia: https://en.wikipedia.org/wiki/1971_Nebraska_vs._Oklahoma_football_game; Last Word on Sports, “The Game of the Century” (Sept. 14, 2021): <https://lastwordonsports.com/collegefootball/2021/09/14/the-game-of-the-century/>.

¹⁰“David M. Nelson,” Wikipedia: https://en.wikipedia.org/wiki/David_M._Nelson (Maine 1949–1950, where he began developing the Wing-T, and Delaware 1951–1965); “Tubby Raymond,” Wikipedia: https://en.wikipedia.org/wiki/Tubby_Raymond (Delaware head coach 1966–2001); Harold R. Raymond, *The Delaware Wing-T: An Order of Football* (Prentice Hall, 1986).

¹¹“LaVell Edwards,” Wikipedia: https://en.wikipedia.org/wiki/LaVell_Edwards (BYU head coach 1972–2000; 1984 national championship); “1984 BYU Cougars football team,” Wikipedia: https://en.wikipedia.org/wiki/1984_BYU_Cougars_football_team; “Ty Detmer,” Wikipedia: https://en.wikipedia.org/wiki/Ty_Detmer (1990 Heisman Trophy). On BYU’s passing game as an equalizer, see Tim Layden, *Blood, Sweat and Chalk* (2010), the BYU chapter.

¹²“Mike Holmgren,” Wikipedia: https://en.wikipedia.org/wiki/Mike_Holmgren (BYU quarterbacks coach before joining the 49ers in 1986); “Andy Reid,” Wikipedia: https://en.wikipedia.org/wiki/Andy_Reid (BYU offensive lineman); “Norm Chow,” Wikipedia: https://en.wikipedia.org/wiki/Norm_Chow.

The Air Raid: subtraction as a strategy

Hal Mumme was the head coach at Iowa Wesleyan, an NAIA school in Mount Pleasant, Iowa, when he hired a lawyer-turned-coach named Mike Leach as his offensive coordinator in 1989. Their plan, borrowed openly from BYU and then pushed much further, was to throw on almost every down from four- and five-receiver sets, with very wide line splits, and to run **very few plays** practised an enormous number of times. Someone brought an air-raid siren to home games to sound after touchdowns, and the offense had its name. The pair took it to Valdosta State (1992–1996) and then to Kentucky, where in 1998 quarterback Tim Couch became a consensus All-American and, the next spring, the first pick in the NFL draft.¹³

Offensive Systems explains the Air Raid's *language* (one-word calls, a slim playbook) and Dropback Concepts draws its signature concepts: mesh, the shallow cross, Y-cross and four verticals. What matters for this chapter is **why a small menu was a weapon**:

- **Repetition beats variety** when your players are worse. A quarterback who has read the same mesh concept against every coverage ten thousand times throws it before the defender has finished deciding.
- **A small menu travels.** It can be taught quickly to a new quarterback every two or three years, and it fits on a wristband or in a hand signal, which makes the no-huddle easy.
- **Spreading five receivers shrinks the box**, so even a pass-first team can run the ball against a light front (more on that arithmetic in the next section).

Leach became Texas Tech's head coach in 2000 and spent a decade there, then Washington State and Mississippi State, until his death in December 2022. His tree became the Air Raid's distribution network: Dana Holgorsen, Sonny Dykes, Kliff Kingsbury and Lincoln Riley among many others.¹⁴ Riley's Oklahoma and USC offenses produced three Heisman-winning quarterbacks who went first in the NFL draft: Baker Mayfield (2017 Heisman), Kyler Murray (2018) and Caleb Williams (2022).¹⁵

i Film room: Texas Tech vs. Texas, November 1, 2008

Leach's best team, quarterback Graham Harrell and receiver Michael Crabtree, beat top-ranked Texas 39–33 in Lubbock when Crabtree caught a pass near the sideline, kept his feet and scored with one second left.¹⁶

What to notice: count the plays. Texas Tech's offense ran a handful of concepts all night from spread formations with very wide line splits, so the Texas defensive linemen had to rush a long way around blockers who were spaced out to give the quarterback clear throwing lanes. And watch where the ball went: short and quick to the sideline, again and again, to receivers who caught it and ran, until the defense crept up and the vertical route behind them came open.

The Air Raid's route into the NFL was, again, people. Patrick Mahomes played for Kingsbury at Texas Tech; Kingsbury became the Cardinals' head coach in 2019, the first NFL head coach from Leach's Air Raid tree, and later Washington's offensive coordinator with Jayden Daniels (04-08 has that film room). Mesh, the shallow cross and four verticals are now in every NFL playbook. What the NFL left at the border was the *pure* system: the very wide line splits that NFL pass rushers would knife through, the near-total absence of a power run game, and the refusal to go under center.

The spread offense: a philosophy, not a formation

By the late 1990s these lineages were converging on one idea, and it's worth naming exactly, because the word gets used loosely. A **spread offense** lines up in the shotgun with three or four receivers spread across the field so that the defense must send defenders out of the box to cover them, then attacks the space that leaves, with quick passes, with option reads, or with both. Spread and Modern Formations drew the distinction: a *spread formation* is a picture every NFL team uses; a *spread offense* is the whole system built on that picture.

The core of the philosophy is arithmetic, the hat math of the run game applied to the whole field.

¹³"Air raid offense," Wikipedia: https://en.wikipedia.org/wiki/Air_raid_offense (Iowa Wesleyan, 1989; Valdosta State, 1992–1996; Kentucky, 1997–1998; Leach on incorporating "much of the BYU passing attack"; the siren); "Mike Leach (American football coach)," Wikipedia: [https://en.wikipedia.org/wiki/Mike_Leach_\(American_football_coach\)](https://en.wikipedia.org/wiki/Mike_Leach_(American_football_coach)); ESPN, "Mike Leach, the Air Raid offense and the return to Kentucky": https://www.espn.com/college-football/story/_/id/30053200/mike-leach-air-raid-offense-return-kentucky.

¹⁴*Sports Illustrated*, "From His Air Raid Offense to His Eccentric Personality, Mike Leach Was One-of-a-Kind" (Dec. 13, 2022): <https://www.si.com/college/2022/12/13/mike-leach-death-legacy-air-raid-college-football>.

¹⁵"Lincoln Riley," Wikipedia: https://en.wikipedia.org/wiki/Lincoln_Riley (Mayfield, 2017 Heisman, first pick in 2018; Murray, 2018 Heisman, first pick in 2019; Williams, 2022 Heisman at USC, first pick in 2024).

¹⁶"2008 Texas vs. Texas Tech football game," Wikipedia: https://en.wikipedia.org/wiki/2008_Texas_vs._Texas_Tech_football_game.

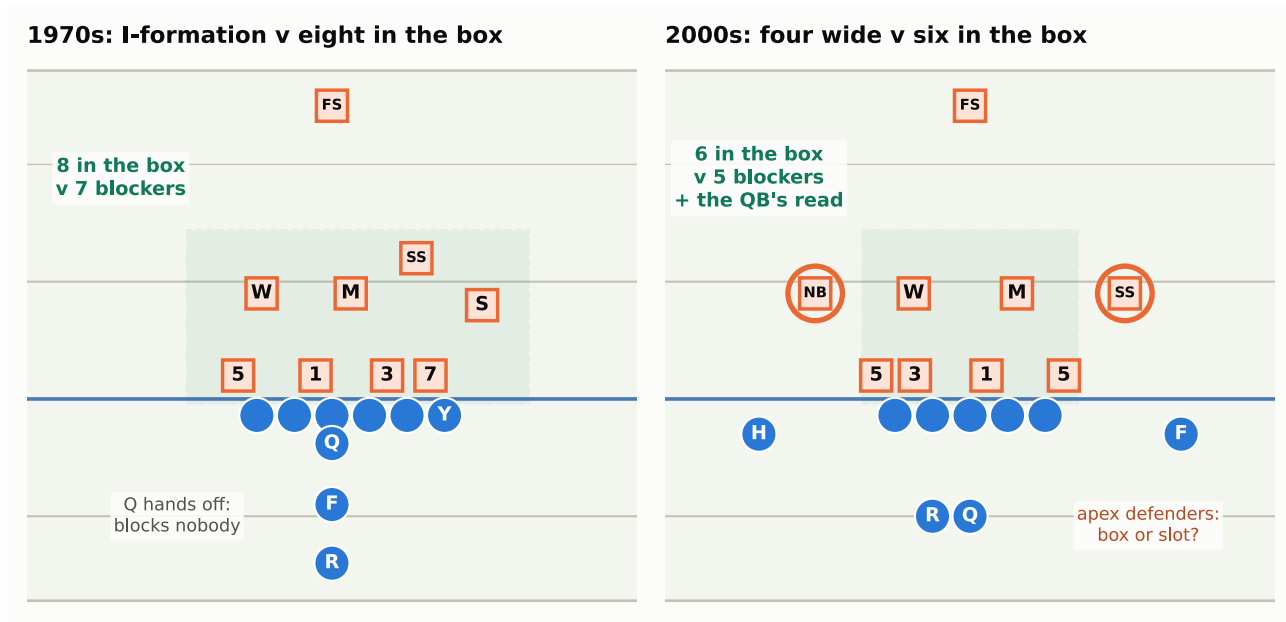


Figure 4: Why spread out to run? The box arithmetic in two eras. Left, a 1970s-style I-formation (21 personnel: a fullback, a tight end and two receivers) against a 4-3 with the strong safety down: eight defenders in the box (shaded) against seven blockers (five linemen, the tight end and the fullback), and the quarterback, who just hands off, blocks nobody. One defender is always free. Right, a four-receiver spread (10 personnel) against a nickel defense: the four receivers have pulled five defenders out wide, two of them (NB and SS, ringed) apex defenders halfway between the box and the slot receivers. Six in the box against five linemen, and the quarterback, by reading one of them, makes it six against six. If either apex defender steps into the box to make it seven, the slot receiver outside him is uncovered. (Wide receivers and cornerbacks are outside both pictures.)

That is the whole idea in one picture. **Spreading out is a way to run the ball**, not only a way to throw it: four receivers take five defenders out of the box, and a defense that brings one back in leaves a receiver alone. The spread then needs one more thing to finish the job, because six defenders against five blockers is still one free man. Two answers emerged, and they split the spread into two families:

- **Spread to pass** (the Air Raid, Joe Tiller's Purdue offense with Drew Brees, the "basketball on grass" offenses of the late 1990s): accept the free man and throw quickly to whoever the defense left alone.
- **Spread to run** (Rich Rodriguez, Urban Meyer and their heirs): make the quarterback a runner, so the read takes care of the free man. That is the spread option, next.

Common misconception

"The spread is a passing offense." Some spread offenses are, but the most influential ones were run-first. Rich Rodriguez's West Virginia, Urban Meyer's Florida and Gus Malzahn's Auburn won with the run game, using the spread to empty the box and the quarterback's legs to win the numbers. The NFL data later in the chapter makes the same point: the shotgun crossed into the NFL first on passing downs, and only later, as the spread run game came with it, on running plays.

The spread option: Rodriguez, Meyer and the quarterback's legs

The **spread option** is a spread offense whose run game is built on option reads from the shotgun. Its origin story is one of the best in football, and 03-04 tells it: at Glenville State in 1991, Rich Rodriguez's quarterback fumbled a zone-play handoff in practice, picked the ball up, saw the backside end crashing down on the running back, and ran where the end had been.¹⁷ Rodriguez saw the mistake was a play. The zone read married the option's read to the most common run in football, inside zone, and to the shotgun, where the quarterback could see the read key and the receivers at once.

Rodriguez took it to Tulane and Clemson as an offensive coordinator, then to West Virginia as head coach (2001–2007), where quarterback Pat White and running back Steve Slaton made it the most dangerous running offense in the country. Urban Meyer,

¹⁷"Zone read still giving defenses fits," Associated Press (Sept. 25, 2014), via *Sports Illustrated*: <https://www.si.com/college/2014/09/25/ap-fbc-zone-read>. Also cited in 03-04.

at Bowling Green, Utah and Florida, built his own spread option on the same principles, with bigger quarterbacks who could run between the tackles: Alex Smith, whose unbeaten 2004 Utah team won the Fiesta Bowl before Smith went first in the 2005 draft, and Tim Tebow, the 2007 Heisman winner, whose Florida teams won national titles in the 2006 and 2008 seasons.¹⁸

What made the spread option more than the zone read was what Rodriguez *attached* to it. With four receivers spread out, the defense's apex defenders (the ones ringed in Figure 4) are stuck between the box and the slots. Rodriguez's quarterbacks read **two** defenders on the same side: the end, to give or keep, and then the apex defender, to keep or throw a bubble screen to the slot. It is the triple option's logic with the pitch replaced by a pass.

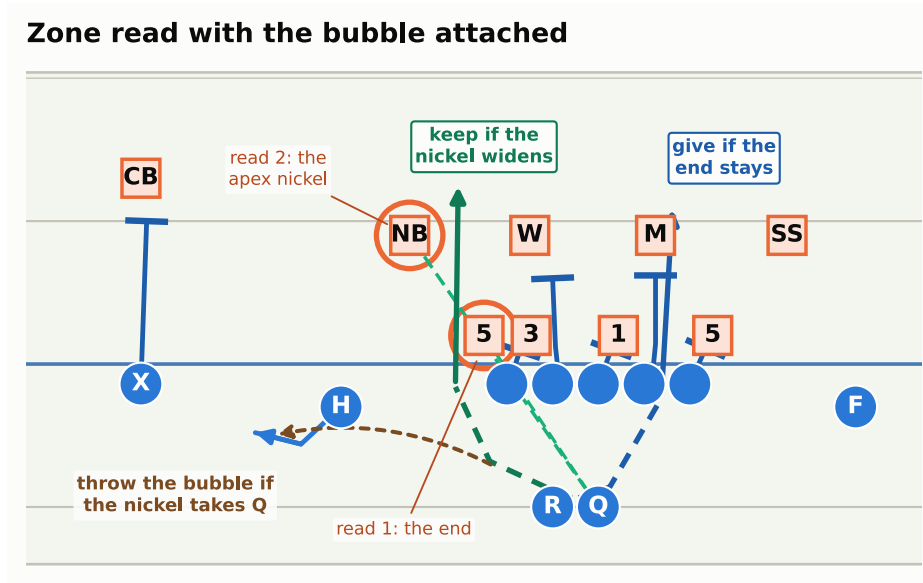


Figure 5: The spread option's three-way play, as Rodriguez's West Virginia teams ran it: the zone read with a bubble screen attached. The line zones to the right and leaves the left end (ringed) unblocked; the quarterback reads him first (dashed green line). If the end stays wide, the quarterback gives (dark dashed arrow) and the back runs the inside-zone track. If the end crashes, the quarterback pulls the ball and reads the second defender on that side, the apex nickel back (ringed), who has to choose between the quarterback and the slot receiver. If the nickel comes up to take the quarterback, the quarterback throws the bubble (brown arc) to H, with X blocking the cornerback; if the nickel widens with H, the quarterback keeps (green dashed arrow) into the space the end and the nickel left. Two reads, three possible ball carriers: the triple option from the shotgun, with a pass as the third option.

Here are the two branches that come out after the end crashes, in motion.

¹⁸"Urban Meyer," Wikipedia: https://en.wikipedia.org/wiki/Urban_Meyer (Bowling Green 2001–2002, Utah 2003–2004, Florida 2005–2010; Utah's 12–0 2004 season and Fiesta Bowl; Florida's 2006 and 2008 titles); "Tim Tebow," Wikipedia: https://en.wikipedia.org/wiki/Tim_Tebow (2007 Heisman); "Alex Smith," Wikipedia: https://en.wikipedia.org/wiki/Alex_Smith (first pick, 2005 draft). On Rodriguez's career, "Rich Rodriguez," Wikipedia: https://en.wikipedia.org/wiki/Rich_Rodriguez.

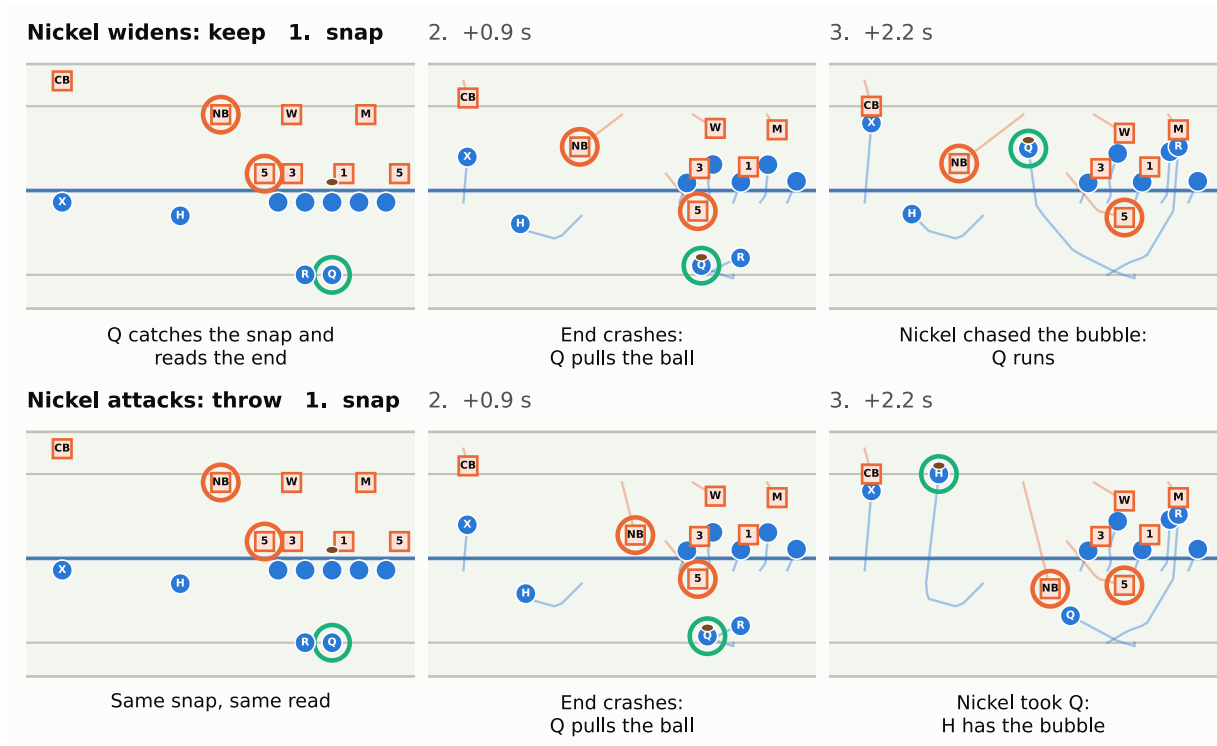


Figure 6: The zone read with the bubble attached, two branches of the same play (simulated, Big Data Bowl coordinates; orange rings mark the two read defenders, the green ring follows the ball; only the left side of the field is shown). In both, the end crashes on the running back, so the quarterback pulls the ball (frame 2). Top row: the nickel widens with the slot receiver's bubble, so the quarterback keeps and runs into the space both defenders left. Bottom row: the nickel comes up to tackle the quarterback, so the quarterback throws the bubble to H, who catches it behind X's block on the cornerback. The defense had two men on that side for three threats.

i Film room: West Virginia, Sugar Bowl, January 2, 2006

Rodriguez's West Virginia, with a freshman quarterback, Pat White, and a freshman running back, Steve Slaton, beat Georgia 38–35 in the Sugar Bowl, played that year in Atlanta while New Orleans recovered from Hurricane Katrina. Slaton ran for 204 yards.¹⁹

What to notice: on the zone reads, find the backside end first, then the apex defender over the slot. Georgia's defense, built to beat the power-running SEC, had fast, big defenders, and it still spent the night a step late, because every snap asked two of them to make a decision they usually didn't get to practise.

Pistol, tempo and the RPO: the 2000s laboratory

Three more inventions finished the modern college offense, and each one came from a coach looking for an edge the others didn't have.

The pistol (Nevada, 2005). Chris Ault put his quarterback about four yards behind the center, between under-center and shotgun depth, with the running back directly behind him. The pistol (02-02) kept the shotgun's view for the quarterback's read while letting the back run downhill to either side, so the defense couldn't tell the play's direction from his alignment. In 2009 Nevada had three 1,000-yard rushers, one of them the quarterback, Colin Kaepernick.²⁰

Tempo (Oregon, Auburn and many others). Chip Kelly, Oregon's offensive coordinator from 2007 and head coach from 2009 to 2012, ran the spread option at a pace college football had never seen: plays signalled from the sideline with picture boards, the

¹⁹"2006 Sugar Bowl," Wikipedia: https://en.wikipedia.org/wiki/2006_Sugar_Bowl.

²⁰"Chris Ault," Wikipedia: https://en.wikipedia.org/wiki/Chris_Ault (the pistol, 2005); Nevada Sports Net, "This week in Pack history: Nevada becomes only NCAA team with three 1,000-yard rushers": <https://nevadasportsnet.com/news/reporters/this-week-in-pack-history-nevada-becomes-only-ncaa-team-with-three-1000-yard-rushers>.

ball snapped as soon as the officials allowed. Kelly went 46–7 as Oregon’s head coach.²¹ Gus Malzahn, a former Arkansas high-school coach who had learned offense from the Delaware Wing-T book, combined tempo, the shotgun and Wing-T misdirection; as Auburn’s offensive coordinator he won the 2010 national title with quarterback Cam Newton (the play we now call the inverted veer was one of their signatures), and as head coach he won the 2013 Iron Bowl on the “Kick Six.”²² In-Game Adjustments and Tempo explains what tempo does to a defense: it freezes the defense’s personnel and its calls, and wears it down.

Why did tempo work so much better on Saturday? Partly the clock: until 2023 the college clock stopped after every first down, so a fast team got even more snaps. Partly the roster: a college team’s backups are much weaker than its starters, so a defense that can’t substitute, or has to play its second string because the first is exhausted, loses more than an NFL defense would. And partly the menu: tempo needs the small, signal-sized playbook the spread already had.

Defenses fought back with the only tool they had left, the rulebook. In October 2012 Nick Saban, whose Alabama defenses were built to substitute by situation, complained about no-huddle offenses and asked, “Is this what we want football to be?”²³ In 2014 the NCAA’s rules committee proposed making offenses wait until 10 seconds had run off the play clock before snapping, so defenses could substitute; after an outcry from coaches, the proposal was tabled before it could be voted on.²⁴ The offense had won the argument.

The RPO (everyone, 2000s–2010s). Put a quick throw on a running play and you have the RPO. Run-Pass Options tells its tangled origin story (Air Raid tags, Rodriguez’s bubbles, Meyer’s accidental discovery at Utah, Brett Favre’s claim to have done it on his own) and why no single inventor can be named. The laboratory conditions explain why it was college that made it routine: the **three-yard** ineligible rule let linemen run-block while the quarterback decided, and the rule’s “crosses the neutral zone” clause made a bubble or a now screen thrown behind the line immune to it entirely. Art Briles’s Baylor teams took the RPO to an extreme, with receivers split almost to the sideline and the quarterback, Robert Griffin III in 2011, reading defenders on the run and the throw on the same snap.²⁵

The defense strikes back

Every offensive idea in this chapter created a problem for defenses, and college defenses, facing a spread offense almost every week, solved those problems first. Five answers matter.

The 3-3-5 stack: speed and mystery

The first answer was simply to match the offense’s speed with defensive backs. The **3-3-5 stack** has three down linemen, three linebackers stacked directly behind them, and five defensive backs, two of whom are hybrids, safeties who play like linebackers, aligned outside the box over the slots. Joe Lee Dunn is usually credited with its first modern version, at Memphis in 1991, born of the familiar college reason: he didn’t have enough big defensive linemen, so he swapped one for a defensive back.²⁶ (Be careful with the name. Odd and Hybrid Fronts explains that NFL coaches use “3-3-5” for something else: a five-man line in nickel, not this stack.)

²¹“Chip Kelly,” Wikipedia: https://en.wikipedia.org/wiki/Chip_Kelly (Oregon offensive coordinator 2007–2008, head coach 2009–2012, 46–7). His NFL tenure is in 08-04.

²²Dennis Dodd, “What’s next for Gus Malzahn and what’s ahead for Auburn after Tigers fire embattled coach,” CBS Sports (Dec. 13, 2020): <https://www.cbssports.com/college-football/news/whats-next-for-gus-malzahn-and-whats-ahead-for-auburn-after-tigers-fire-embattled-coach/>; “Gus Malzahn,” Wikipedia: https://en.wikipedia.org/wiki/Gus_Malzahn (Auburn offensive coordinator 2009–2010, head coach 2013–2020); on the Delaware Wing-T as his model, see the sources in 03-05.

²³CBS Sports, “Nick Saban no fan of no-huddle offenses, asks ‘Is this what we want football to be?’” (Oct. 2012): <https://www.cbssports.com/college-football/news/nick-saban-no-fan-of-no-huddle-offenses-asks-is-this-what-we-want-football-to-be>.

²⁴ESPN, “NCAA committee tables planned vote on 10-second rule proposal” (Mar. 5, 2014): https://www.espn.com/college-football/story/_/id/10556971/ncaa-committee-tables-planned-vote-10-second-rule-proposal.

²⁵“Art Briles,” Wikipedia: https://en.wikipedia.org/wiki/Art_Briles; “Robert Griffin III,” Wikipedia: https://en.wikipedia.org/wiki/Robert_Griffin_III (2011 Heisman). On Baylor’s wide splits and packaged plays, and Mike Shanahan borrowing them for Griffin in 2012, see the sources in 04-06.

²⁶Chris B. Brown, “The Birth of the 3-3-5 Defense,” Grantland (June 6, 2012): <https://grantland.com/features/an-excerpt-essential-smart-football-birth-3-3-5-defense/>. Also cited in 05-03.

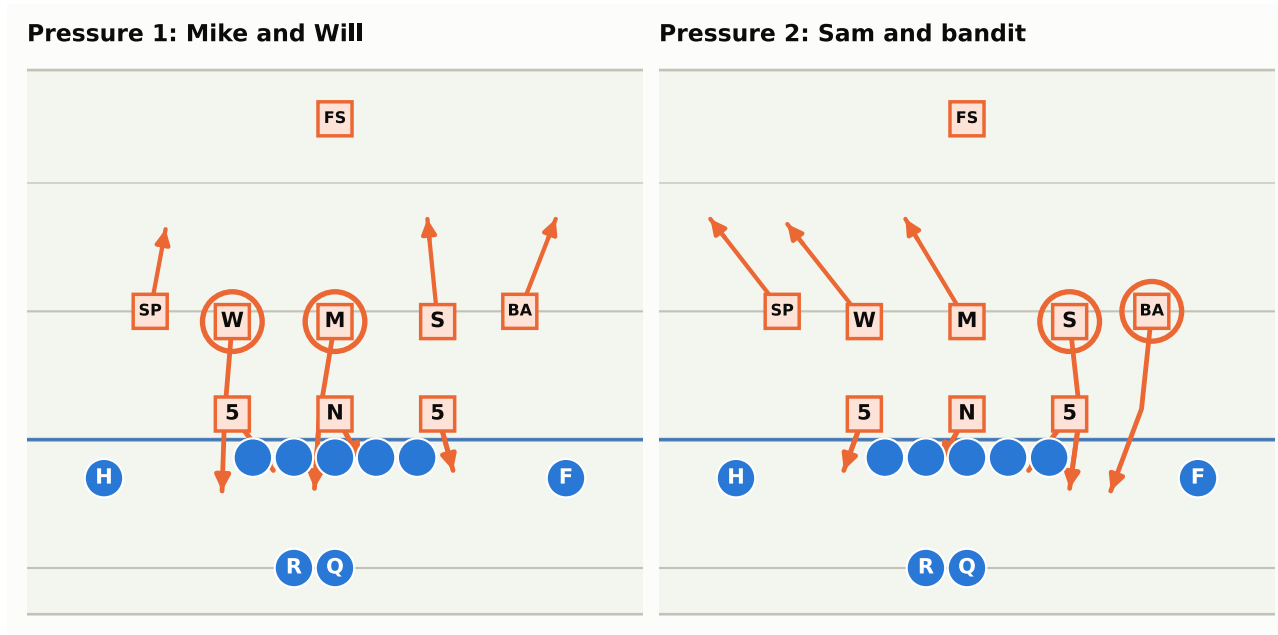


Figure 7: The 3-3-5 stack against a four-receiver spread, and why offenses hated it. Both panels show the same pre-snap picture: three down linemen (a nose on the center and two 5-techniques), three linebackers stacked directly behind them, and two hybrid safeties (SP, the spur, and BA, the bandit) aligned outside the box over the slots, with two cornerbacks (out of the picture) on the wide receivers and one deep free safety. Eight defenders could rush, but only three have their hands down, so the offensive line can't tell before the snap who is coming. Left: the line slants to the offense's right while the Mike and Will blitz through the gaps the nose and left end vacated; five rush, six cover. Right: the line slants the other way, the Sam and the bandit come from the offense's right, and the Mike and Will drop. Same picture, different five rushers, and a protection that blocked the first picture's rushers is wrong for the second.

The stack's weapons are **speed** (an extra defensive back against an extra receiver) and **mystery**: with eight potential rushers standing up, a protection scheme that counts the defenders on the line before the snap counts only three. Its price is size. Three linemen against five offensive linemen can be pushed around, which is why the stack thrived in conferences full of spread passing offenses and struggled against teams that lined up with tight ends and ran straight at it. West Virginia, under Rodriguez's defensive coordinator Jeff Casteel, paired the stack with the spread option, so the Mountaineers practised against the offense they defended.²⁷ The NFL never adopted it as a base defense; NFL offensive lines are too good to give away a lineman every down.

Quarters and pattern matching: the college two-high

The second answer was in the secondary. Against spread offenses that ran option and play-action from the same looks on wide college hashes, a defense that put a safety in the box got beaten deep, and one that kept two safeties deep got run over. **Quarters** with pattern matching let it do both: two safeties 10 to 12 yards deep who read the receivers in front of them, carry the vertical routes and fit the run from depth when it shows. *Quarters and Match Coverage* tells this history in full: Nick Saban's pattern-match rules, developed as Bill Belichick's coordinator in Cleveland and refined at Michigan State, LSU and Alabama; Mark Dantonio and Pat Narduzzi's Michigan State; Iowa's long-running split-field quarters; Gary Patterson's 4-2-5 at TCU.

This is the one lineage that made a round trip. Its roots were in an NFL building (Cleveland in the early 1990s), it matured on college fields against the spread, and when NFL offenses went spread in the late 2010s, NFL defenses went back to the college quarters coaches for the answer. NFL Next Gen Stats measured split-safety coverage on 32.9% of dropbacks in 2018 and 42.0% in 2025, the highest since tracking began.²⁸

Tite and mint: walling off the middle

The third answer was a front built for the spread's favourite run. With five or six defenders in the box, a defense can't afford to let inside zone's double teams climb to the linebackers. The **tite front** puts a nose on the center and two 4-techniques on the tackles'

²⁷CBS Sports, "Why defenses are trying to Stack up with spread offenses dominating throughout college football": <https://www.cbssports.com/college-football/news/why-defenses-are-trying-to-stack-up-with-spread-offenses-dominating-throughout-college-football>.

²⁸NFL.com, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm" (Next Gen Stats): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>. The sources for the Saban, Michigan State, Iowa and TCU history are in 06-05 and 06-06.

inside shoulders, so every interior lineman is covered and nobody climbs cleanly (05-03 draws it). Kirby Smart's Georgia calls the same picture the mint front, and Georgia's defenses won national titles in the 2021 and 2022 seasons with it as a base.²⁹ It is also a case of parallel evolution rather than borrowing: Vic Fangio's NFL defenses were running their own title against the same spread run games, and the two traditions met in the middle.

Three-high: Iowa State's answer to the Air Raid

The newest answer came from the conference where the Air Raid lived. By 2017 the Big 12 was a league of spread passing offenses, and Iowa State, under head coach Matt Campbell and defensive coordinator Jon Heacock, was losing shootouts. Midway through that season Heacock threw out his four-man front and lined up in a 3-3-5 with **three safeties deep**. Within a few weeks of the switch, Iowa State, a 31-point underdog with a backup quarterback, went to Norman and beat third-ranked Oklahoma, Baker Mayfield's team, 38–31.³⁰

A **three-high safety defense** shows three safeties deep before the snap, usually from a 3-3-5. It is built on one bet: **make the offense run the ball against a light box, and fill the box late**.

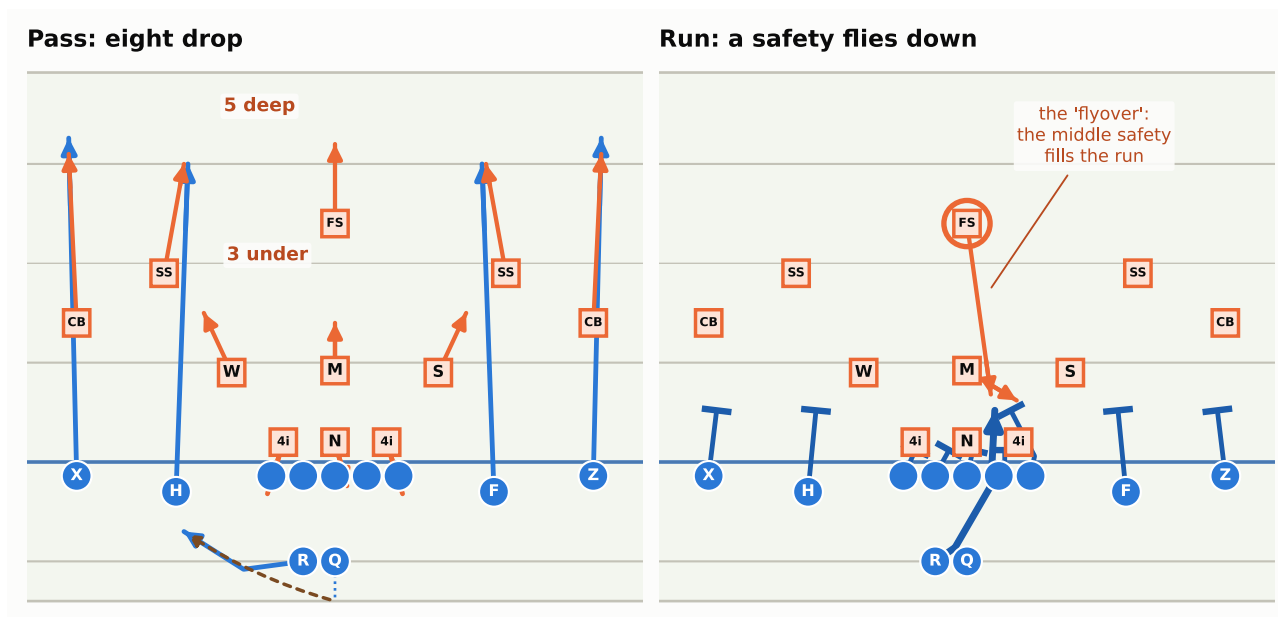


Figure 8: The three-high safety defense against a four-receiver spread, one pre-snap picture, two answers. Before the snap: three down linemen, the Mike in the middle, the Will and Sam walked out over the slots, two cornerbacks, and three safeties (SS, FS, SS) 9 to 12 yards deep (receiver splits drawn a little tighter than usual, to fit the page). Left, against a pass: eight drop into coverage (five deep, three underneath) and three rush, so four vertical routes run into five deep defenders; the quarterback has to settle for a short throw. Right, against a run: the line and the Mike take on inside zone, and the middle safety (ringed) ‘flies’ down into the box after the snap to make the tackle, so a box that looked like five defenders against five blockers before the snap has six or seven by the time the back arrives.

Why does that work against the Air Raid in particular? Because the Air Raid's whole menu is built to find the hole in a two-high or one-high shell: the seam against Cover 3, the deep middle against Cover 2, the crossers underneath against man. Against five deep defenders and three underneath, there is no deep hole, and the offense's best answer, running the ball against a light box, is something many pure passing offenses were least prepared to do. The cost is the run defense if the safety is late: a three-high defense that doesn't fill on time is a five-man box against a five-man line and a running back with a head start. Iowa State's version spread through the Big 12 and beyond (Kansas's offensive coordinator called Iowa State “the originators”), and three-safety looks

²⁹“2021 Georgia Bulldogs football team” and “2022 Georgia Bulldogs football team,” Wikipedia: https://en.wikipedia.org/wiki/2021_Georgia_Bulldogs_football_team, https://en.wikipedia.org/wiki/2022_Georgia_Bulldogs_football_team. On the mint front, see the sources in 05-03.

³⁰CBS Sports, “Iowa State at Oklahoma score: Cyclones shock No. 3 Sooners in major Power Five upset” (Oct. 7, 2017): <https://www.cbssports.com/college-football/news/iowa-state-at-oklahoma-score-cyclones-shock-no-3-sooners-in-major-power-five-upset>; on the mid-season switch, the 31-point spread and how few games the new scheme had been in use, Cyclone Fanatic, “Synergy on ‘O’ and ‘D’ defining the 2017 Cyclones” (Oct. 2017): <https://cyclonefanatic.com/2017/10/gray-synergy-on-o-and-d-defining-the-2017-cyclones/>.

³¹Lawrence Journal-World, “At Iowa State, KU will face defensive scheme that has caused past issues” (Nov. 1, 2023), quoting Kansas offensive coordinator Andy Kotelnicki: <https://www2.ljworld.com/sports/2023/nov/01/at-iowa-state-ku-will-face-defensive-scheme-that-has-caused-past-issues/>; Cody Alexander, *MatchQuarters*, on Heacock's three-high run fits: <https://www.matchquarters.com/p/iowa-state-heacock-3-high-run-fits>.

became a regular part of college defense.³¹ NFL defenses have mostly borrowed the *personnel* (a third safety on the field, the big nickel of A First Look at Defense) and the habit of disguising the shell before the snap (06-06), rather than the full college package.

The defensive coach who switched sides

The best evidence that the spread won the argument came from its loudest critic. Saban's Alabama, the most dominant defensive program of the 2010s, lost games to up-tempo spread teams (including Malzahn's Auburn on the Kick Six and Ohio State's spread offense in the playoff that ended the 2014 season), and Saban answered by hiring spread coordinators, starting with Lane Kiffin in 2014. By 2020 his national champions were an up-tempo, RPO-heavy spread offense that put up points by the bucketful.³²

Crossing over: how the ideas reached Sunday

Now set the lineages against the NFL's filter. The table puts each idea next to its route into the league and what changed on the way.

Table 1: Ten college ideas through the NFL's filter. "Peaked" is approximate; the text and notes give dates and sources.

College idea	Where it peaked	How it reached the NFL	What the NFL kept	What it left behind
Triple option (veer, wishbone, flexbone)	Houston, Texas, Oklahoma, 1965–1985; the academies since	It didn't, as a system	The idea of reading a defender	The hits on the quarterback, the three-back backfield
Wing-T	Delaware, 1950s–1990s; Malzahn's Auburn	Coaches' clinics; the jet-motion era (Rams, 49ers, late 2010s)	Series football, jet motion, buck sweep, the bootleg	The full under-center Wing-T backfield
BYU passing	BYU, 1972–2000	People: McMahon, Young, Holmgren, Reid	Spacing and crossing routes	— (it was an ancestor)
Air Raid	Kentucky, Texas Tech, Oklahoma (Riley), 1997–2020	Quarterbacks (Mahomes, Mayfield, Murray, Williams), then coaches (Kingsbury, 2019)	Mesh, shallow cross, Y-cross, four verticals; no-huddle	Very wide line splits, the near-absence of a run game
Spread option and zone read	West Virginia, Utah, Florida, 2001–2010	Quarterbacks (Griffin, Kaepernick, Wilson, Newton, 2012; Jackson, 2018)	The zone read and its relatives as a package	A run game built <i>entirely</i> on the quarterback
Pistol	Nevada, 2005–2012	Washington and San Francisco, 2012; Baltimore, 2019	The formation, especially for running quarterbacks	—
Tempo	Oregon, Auburn, Baylor, 2009–2015	Chip Kelly's Eagles, 2013	No-huddle as an information tool	Tempo as a whole-game identity (it faded; see 08-04)
RPO	Everywhere, 2005–2015	Reid's Chiefs and Pederson's Eagles, 2016–2018	Quick-throw RPOs	Slower RPOs that need three yards for the linemen
Match quarters	Michigan State, Iowa, TCU, Alabama, 1995–2015	Coaches (Fangio's tree, Staley), late 2010s	Two-high, split-field match coverage	— (it came home)
3-3-5 stack, three-high	Memphis, West Virginia, Iowa State	Mostly hasn't	A third safety in nickel personnel; disguise	Three down linemen as a base

Three patterns run through Table 1.

³²The Ringer, on Saban's evolution from the 2012 no-huddle complaint to a spread offense (Jan. 12, 2021): <https://www.theringer.com/2021/1/12/22226338/alabama-national-championship-nick-saban-process-evolution>.

Players carry ideas better than coaches. The ideas that crossed fastest arrived inside quarterbacks: the zone read with Griffin, Kaepernick and Wilson in 2012, the Air Raid with Mahomes in 2017 and 2018, the designed quarterback run game with Lamar Jackson. NFL coaches built offenses around what their young quarterbacks already knew how to do. College *head coaches* who tried to bring whole systems with them had a much harder time. Steve Spurrier (Washington, 2002–2003), Nick Saban (Miami, 2005–2006), Chip Kelly (Philadelphia, 2013–2015, then San Francisco), Urban Meyer (Jacksonville, 2021) and Matt Rhule (Carolina, 2020–2022) all left the NFL with losing or near-.500 records within three seasons, while ideas from their college offenses and defenses stayed behind.³³ The NFL takes the play and sends back the coach.

The NFL adapts what it keeps. The RPO crossed, but as a *quick-throw* RPO, because the one-yard rule won't let linemen run-block while the quarterback waits. The zone read crossed, but as a few plays a game, because the quarterback test is harsh. The Air Raid crossed, but inside offenses that also go under center, use tight ends and run power. Every idea arrives a little smaller.

The crossing has a rhythm. Ideas arrive in waves when a generation of players makes them unavoidable, and then the NFL's defenses answer and the wave recedes. The zone read surged in 2012 and receded by 2014 (Dynasties and Counterpunches tells that story); the shotgun spread rose for twenty years and has just started to give ground to the under-center game (The Modern Era). Here is that long wave in data.

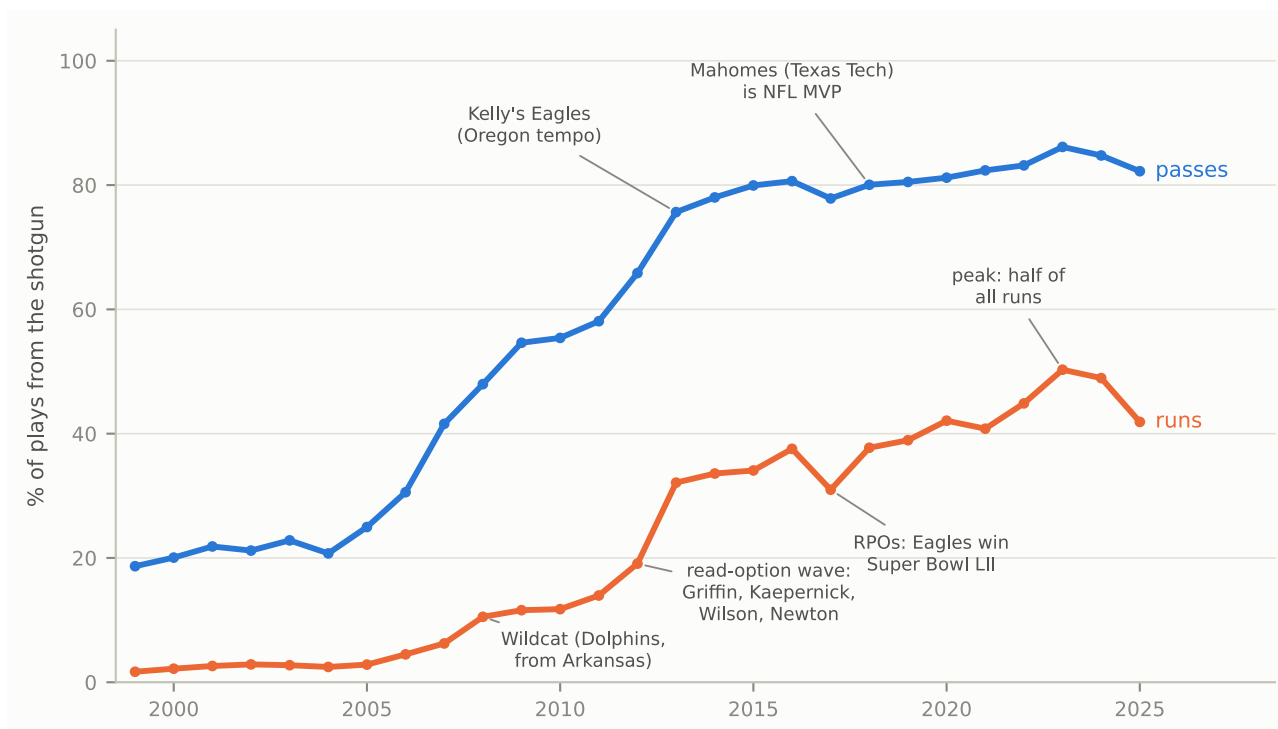


Figure 9: The spread's long arrival: the share of NFL plays snapped from the shotgun, 1999–2025 (regular season), shown separately for passes (dropbacks, scrambles included) and runs, with some of the college imports of the text marked. The passing game moved to the shotgun first; the run game followed only once offenses learned to run from it, with zone reads, the pistol and RPOs. Shotgun runs went from about 2% of runs in 1999 to half of them in 2023, before the under-center revival of 2024–2025. The pistol is counted as shotgun in this data. Data: nflverse play-by-play via nflreadpy (the [shotgun](#) flag).

Read Figure 9 as two stories. The **passing** line is the spread formation arriving: from about 19% of dropbacks in 1999 to 55% in 2009, as offenses learned what college coaches had learned, that a quarterback who starts five yards deep sees the defense better and gets the ball out faster. That shift needed no college import at all; the 2007 Patriots, the most famous spread passing offense of its era, were run by NFL lifers. The **running** line is the college story. In 2011, only 14% of NFL runs came from the shotgun: the NFL used the gun to pass and went under center to run. Then the read-option wave and Chip Kelly's Eagles arrived, and shotgun runs jumped to 32% in 2013. The spread run game, the zone read, the pistol and then the RPO, had crossed. By 2023 the shotgun accounted for 50% of runs and 72% of all plays. In 2025 both lines fell back (to 66% of all plays), as NFL offenses went back under center to run and play-action, the latest counterpunch, told in The Modern Era. Two cautions: the play-by-play's [shotgun](#) flag

³³NFL head-coaching records from Pro Football Reference's coach pages: Spurrier 12–20; Saban 15–17; Kelly 26–21 with the Eagles (released before the final game of 2015) and 2–14 with the 49ers in 2016; Meyer 2–11 (fired during the 2021 season); Rhule 11–27 (fired during the 2022 season): <https://www.pro-football-reference.com/coaches/>.

comes from the official gamebook descriptions, which probably recorded it less consistently in the early seasons, and it can't tell a spread-option shotgun run from a plain shotgun handoff.³⁴

Saturday versus Sunday: the rules that shape each game

You can't watch a college game well without the rulebook differences, because many of the strategies in this chapter exist *because* of them. Here they are side by side, current for the 2026 seasons. (The full reference version lives in Appendix E.)

³⁴This book's calculation from nflverse play-by-play (`load_pbp`), regular seasons 1999–2025, plays with `play_type` “pass” or “run” and a down; scrambles counted as passes; `shotgun` is the play-by-play flag (pistol snaps are included). It matches the shotgun shares in this book's fact sheet for 2016–2025.

Table 2: The college and NFL rules that most change strategy, as of the 2026 seasons. Sources in the notes below the table's discussion.

Rule	NFL (2026)	College (2026)	What it changes on the field
Hash marks	70 ft 9 in from each sideline; 18 ft 6 in apart	60 ft from each sideline; 40 ft apart	College has a true wide side and short side on most snaps; formations and play calls are built around it
Catch at the sideline	Two feet (or another body part) in bounds	One foot in bounds	Sideline throws are easier to complete in college
Ineligible linemen downfield on a pass	No more than 1 yard, on any forward pass	No more than 3 yards, and only on passes that cross the neutral zone	College RPOs can run-block longer and throw deeper
Defensive pass interference	Spot foul, automatic first down	15 yards from the previous spot (or the spot, if shorter), automatic first down	A deep NFL throw into tight coverage can "win" 40 yards on a flag; in college the flag is capped
Offensive pass interference	10 yards	10 yards (15 until 2025)	Cut for 2026, matching the NFL
Clock after a first down	Keeps running	Stops only in the last two minutes of each half (stopped all game until 2023)	Pre-2023 college games had many more snaps; tempo offenses profited
Two-minute warning	Yes	Yes, since 2024 (the "two-minute timeout")	Late-half clock strategy now looks similar
Ball carrier down	When touched by an opponent while on the ground	As soon as a knee or other body part touches the ground, touched or not	No "get up and keep running" in college
Overtime	One 10-minute period in the regular season (15 in the playoffs); both teams get the ball; then sudden death	Alternating possessions from the 25; a two-point try is required after a touchdown from the second overtime; alternating two-point plays from the third	College overtime is a series of short-field possessions; strategy becomes two-point math
Targeting / hits to the head	15 yards; ejection only for flagrant hits (use-of-helmet rule since 2018)	15 yards and automatic ejection; for 2026, a first offense no longer carries a suspension into the next game	College defenders must change their tackling target or lose a starter
Kickoffs	The "dynamic kickoff" (since 2024), built to be returned	Traditional kickoff; a fair catch inside the 25 is a touchback at the 25	College kickoffs are mostly fair catches and touchbacks
Fair-catch kick	Allowed for decades (rarely used)	New for 2026	A free 3-point try after a fair catch near midfield, now in both games
Motion	One player, not moving forward at the snap	The same	Differences you see in motion are usage, not law
Helmet communication	Quarterback and one defender (green dot) since the 1990s and 2008	One player per side, since 2024	College offenses used sideline signals and picture boards for decades, which is part of why tempo grew there

A few of these deserve a sentence on *why* they matter.

The hash marks are the one you can see from your couch. With the ball on a college hash, offenses put their strength into the wide field and leave a single receiver into the short side, and defenses set their coverage by the field and the boundary rather than by the formation. That is why college defenses grew field-and-boundary coverages (and split-field quarters, 06-06), and why college kickers sometimes waste a down centering the ball before a field goal.³⁵

The three-yard rule, the one-foot catch and the capped DPI all favour offenses that throw in rhythm off the run and to the sideline. Together they are the legal foundation of the college RPO.³⁶ The NFL's spot-foul DPI pushes the other way: it makes a deep throw into tight coverage a reasonable gamble, a strategy Contact Rules and Penalties measures.

The college clock used to stop after every first down, which added roughly seven snaps a game and made the up-tempo offenses of the 2000s and 2010s even more potent. The NCAA changed it for 2023, so the clock now keeps running after a first down except in the last two minutes of each half, and added a two-minute timeout for 2024.³⁷ The college game has quietly moved toward the NFL's.

Overtime is a different sport in college. Since 2021, teams alternate possessions from the opponent's 25, must go for two after a touchdown from the second overtime, and from the third overtime simply alternate two-point plays.³⁸ The NFL's version is in Clock and Game Management.

Targeting has made the college defender's aiming point a strategic question. A targeting foul ejects the player; for 2026, in a one-year trial, a player's first targeting foul of the season no longer costs him part of the next game, but second and third fouls still do.³⁹ The NFL has no automatic ejection for helmet contact.

Kickoffs have gone in opposite directions. The NFL rebuilt its kickoff in 2024 to bring the return back (Kickoffs and Returns), while college, since 2018, has let the receiving team fair-catch a kickoff inside its 25 for a touchback at the 25, so most college kickoffs are not returned.⁴⁰ For 2026 the NCAA also adopted the fair-catch kick, the old NFL rule that lets a team attempt a free field goal from the spot of a fair catch.⁴¹

Helmet communication came to college football only in 2024: one player per side with a speaker in his helmet, cut off at 15 seconds on the play clock, as in the NFL.⁴² For the decades before that, college teams called plays with signals and giant picture boards, which is one reason the no-huddle, signal-sized playbooks of the spread grew up there: an offense that already signalled every play lost nothing by going fast.

³⁵On centering the ball before college field goals, see 09-01; on field-and-boundary coverage, 06-06.

³⁶NCAA Football Rules (2026): Rule 7-3-10 (three yards; see the note above), Rule 2-4-3 (a catch requires one foot in bounds), Rule 7-3-8 (pass interference: defensive, 15 yards or the spot; offensive, 10 yards for 2026). National Football Foundation summary of 2026 rule changes, reported in the *Altoona Mirror* (Aug. 11, 2026): <https://www.altoonamirror.com/sports/psu/2026/08/cfb-nff-rules-targeting-penalty-changing/>; the NFL's spot-foul rule and the college comparison are in 09-03.

³⁷NCAA.com, "Fewer clock stoppages on first downs and more 2023 college football rule changes" (Aug. 25, 2023): <https://www.ncaa.com/news/football/article/2023-08-25/fewer-clock-stoppages-first-downs-and-more-2023-college-football-rule-changes> (the clock stopped on first downs from 1968 to 2022; the change was expected to cut about seven plays a game); CBS Sports, "NCAA approves 2-minute warning in college football games, helmet communications beginning in 2024 season" (Apr. 2024): <https://www.cbssports.com/college-football/news/ncaa-approves-2-minute-warning-in-college-football-games-helmet-communications-beginning-in-2024-season>.

³⁸NCAA, "New football overtime rules displayed this season" (Oct. 27, 2021): <https://www.ncaa.org/news/2021/10/27/general-new-football-overtime-rules-displayed-this-season.aspx>. The NFL regular-season overtime rule of 2025 is summarized in this book's fact sheet and in 10-04.

³⁹NCAA, "Changes to penalty structure for targeting in DI football approved" (Mar. 19, 2026): <https://www.ncaa.org/news/2026/3/19/media-center-changes-to-penalty-structure-for-targeting-in-di-football-approved.aspx>; the NFL's use-of-helmet rule (2018) is in 09-03.

⁴⁰The NCAA's 2018 kickoff rule: a fair catch of a kickoff inside the receiving team's 25 is a touchback at the 25. "Kickoff (gridiron football)," Wikipedia: [https://en.wikipedia.org/wiki/Kickoff_\(gridiron_football\)](https://en.wikipedia.org/wiki/Kickoff_(gridiron_football)).

⁴¹NCAA, "Changes to penalty structure for targeting in DI football approved" (Mar. 19, 2026), which also lists the fair-catch kick and the 10-yard offensive pass interference penalty for 2026 (as above); Bleacher Report, "NCAA committee proposes rule changes: targeting suspensions, OPI penalties, fair catch kick and more": <https://www.bleacherreport.com/articles/25401192-ncaa-committee-proposes-rule-changes-targeting-suspensions-opi-penalties-fair-catch-kick-more>.

⁴²NCAA, "Technology rules approved in football" (Apr. 19, 2024): <https://www.ncaa.org/news/2024/4/19/media-center-technology-rules-approved-in-football.aspx> (one player per team, identified by a green dot, cut off at 15 seconds on the play clock or the snap). On the NFL's helmet radio, see 01-04.

Madden vs. real life

In a video game, every team in the league has roughly the same playbook menus, and so does every team in the college game. Real college football is closer to a hundred different sports. On the same Saturday you can watch a service academy run the flexbone triple option, an Air Raid team throw 60 times from five-receiver sets, and an SEC power program line up under center with two tight ends. NFL teams are far more alike than that, because they draw from the same talent pool under the same rules and copy each other within a season. That variety is the laboratory. It's also why a college game is a good place to see a concept in its purest form: the team that runs it is often the only one that does.

Watch for it

Saturday versus Sunday. Watch one college game and one NFL game on the same weekend, and list three structural differences you can see. Some places to look:

Before the snap

- **Where is the ball?** In college, on most snaps it sits on a hash 20 yards from one sideline. Which way does the offense put its receivers: into the wide field or the short side? Does the defense line up by the field and the boundary, or by the formation?
- **How fast is the next snap?** Time a few series from whistle to snap. College tempo teams snap in 10 to 15 seconds; how many NFL teams do?
- **Who is in the backfield?** Count the formations with the quarterback under center. Then count the option looks (a back beside the quarterback, an unblocked end) and how often the quarterback actually keeps.
- **How many safeties are deep?** Look for college defenses with three safeties back, and for three-down-lineman fronts with stacked linebackers.

After the snap

- **On a run with a throw attached,** watch the offensive linemen: in college they may be two or three yards downfield when the ball is thrown. In the NFL, watch for the quick throw that comes out before they can get there.
- **On a sideline catch,** count the feet. One in college, two in the NFL.
- **On a long pass-interference flag,** where does the ball go? Fifteen yards in college; the spot of the foul in the NFL.

For your prediction log (01-06): on every college option or RPO look, write who the quarterback is reading, and how many defenders are in the box against how many blockers.

Predict the play

💡 Predict the play

Drill 1: count the Saturday box. First-and-10 at midfield in a Big 12 game. The ball is on the left hash. The offense lines up in a four-receiver spread with the running back beside the quarterback. The defense shows three down linemen, a Mike linebacker over the ball, the Will and Sam walked out over the slots, and three safeties deep. Count the box against the blockers. Is the offense more likely to run or pass, what kind of play, and what is the defense counting on?

Drill 1: 1st and 10, ball on the left hash

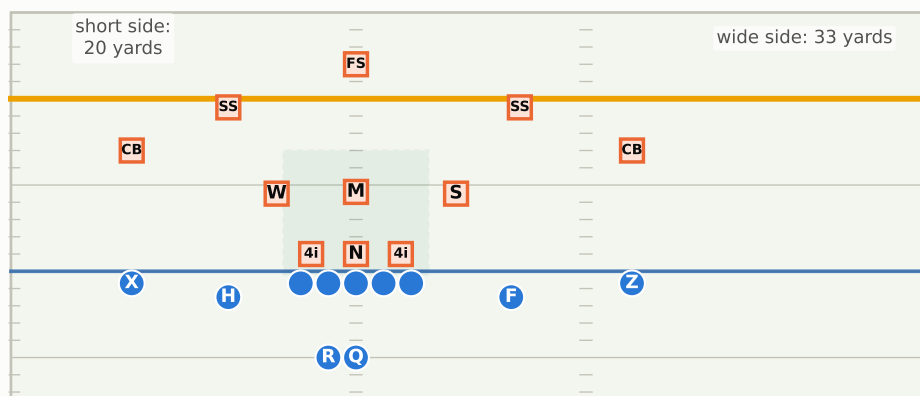


Figure 10: Drill 1: a four-receiver spread (10 personnel) against three-high, on a college field with the ball on the left hash (the hash marks are drawn at college width). The shaded strip is the box.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2: the wishbone count. A wishbone team runs the triple option to its split-end side (the offense's left). On the first four snaps, the defense's left end (the 5-technique) has squeezed down hard on the fullback every time, and the outside linebacker standing on the line outside him has run straight at the pitch man every time. Before the next snap, name #1, #2, #3 and #4, and predict who will carry the ball.

Drill 2: who carries it?

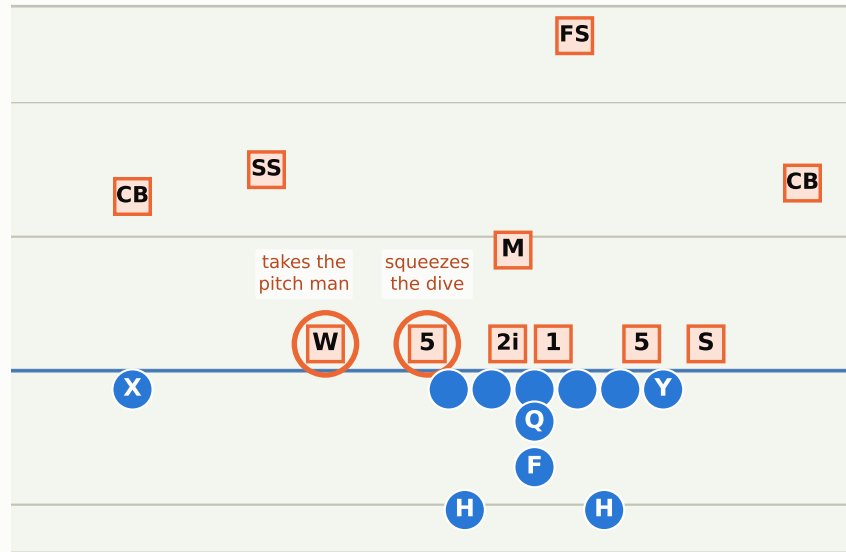


Figure 11: Drill 2: wishbone, triple option to the left. The two defenders who have been doing the same thing every snap are ringed.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3: one play, two rulebooks. An offense runs an RPO: the line run-blocks inside zone, and the quarterback pulls the ball and throws to the slot receiver on a seam route that bends to the sideline. He catches it 17 yards downfield, right at the sideline, with one foot down before stepping out. At the moment of the throw, the right guard, who came off a double team toward a linebacker, is 2½ yards past the line of scrimmage, still not touching anyone. What is the ruling in a college game? In an NFL game?

Drill 3: legal on Saturday? On Sunday?

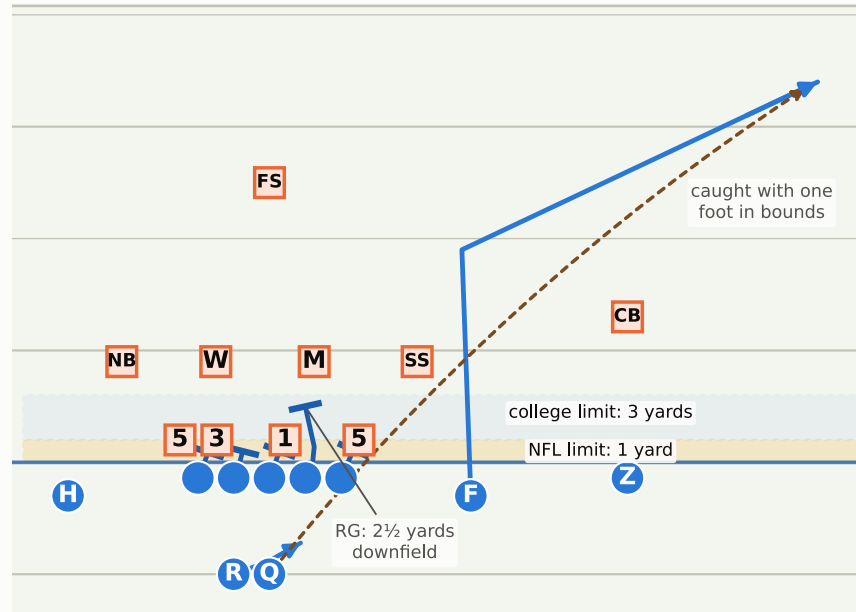


Figure 12: Drill 3: the RPO at the moment of the throw. The shaded strips mark 1 and 3 yards past the line of scrimmage; the right guard's block ends between them. The slot receiver catches the ball at the sideline.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- College football is the sport's laboratory because of its environment: **huge talent gaps** across hundreds of programs, **wide hash marks**, rules like the **three-yard** ineligible limit that tolerate the run-pass blur, and **roster churn and tempo** that reward small, repeatable systems.
- The offensive lineages share one aim, changing the arithmetic for the team with lesser players: the **option** (split-T, veer, wishbone, flexbone, spread option) reads a defender instead of blocking him; the **Wing-T** makes him guess; **BYU and the Air Raid** spread the ball to five receivers and repeat a few plays endlessly; the **spread offense** empties the box so the run game has numbers.
- The **spread option** is the triple option rebuilt from the shotgun, with the pitch replaced by a throw; add the three-yard rule and tempo and you have the modern college RPO offense.
- Defenses answered with **speed and disguise** (the 3-3-5 stack), **two-high match coverage** (quarters, which made a round trip back to the NFL), **fronts that wall off inside zone** (tite and mint) and **three-high safeties** that concede the light-box run and fill it late.
- The NFL is a filter: ideas that hit the quarterback, rely on a talent gap or need college rules arrive smaller or not at all. **Players carry ideas across better than coaches**, which is why the big crossings (2012's read option, 2017's RPOs, the Air Raid's concepts) came with young quarterbacks.

Connections

Builds on: 1978 and the Passing Revolution (the NFL side of the same decades), Option Football (the reads behind the option lineage), Run-Pass Options (the RPO and the one-yard rule) and Quarters and Match Coverage (the defensive answer that made

the round trip). The formations are in Classic Formations and Spread and Modern Formations, and the Air Raid's language and concepts in Offensive Systems and Dropback Concepts.

Sets up:

- Dynasties and Counterpunches (2000–2016): the 2012 read-option wave and why it receded, and Chip Kelly's Eagles.
- The Modern Era: two-high defenses, the under-center comeback, and the NFL's latest borrowing from Saturday.
- Case studies: the Ravens' option run game (the spread option's fullest NFL expression), Andy Reid's Chiefs (RPOs and an Air Raid quarterback), the Eagles (RPOs and the quarterback run game) and Fangio's two-high revolution.
- Appendix E: the rules table and the coaching trees.

If you want to go deeper: Chris B. Brown's *The Art of Smart Football* and *The Essential Smart Football* (the zone read, the Air Raid and the 3-3-5's birth, among much else); Tim Layden's *Blood, Sweat and Chalk*, whose chapters on the wishbone, the veer, BYU, the zone read and the spread are told largely in the coaches' own words; S. C. Gwynne's *The Perfect Pass*, on Mumme, Leach and the Air Raid; and Cody Alexander's *MatchQuarters* writing on modern college defense, including Iowa State's three-high.

Answers to Predict the play

Answer 1.

Count it: five in the box against five blockers. Three linemen and the Mike are inside the box, and the Will is right on its edge over the short-side slot: call it five, with a sixth (the Sam) out over the slot to the wide side. The offense has five linemen, and the quarterback can read one defender, so on a zone read the offense has a blocker or a read for every man in the box, with the running back as the extra runner. **Expect a run, most likely a read** (a zone read or an RPO that reads the Will), because the defense has invited it.

What the defense is counting on is the three safeties: on a run, one of them, usually the middle one, comes down after the snap to fill, so the box the offense counted before the snap gains a defender by the time the back gets there (Figure 8). That's the bet of the three-high: give the offense a light box, take away everything deep, and tackle the run late. If the safety is a step slow, the run is a big gain; if he's on time, the offense has run into a seven-man box it didn't see.

Bonus: why you knew it was college. The ball is 20 yards from the left sideline, not 23.6: college hash marks. And the three-safety shell is a college look.

Answer 2.

#1 is the 5-technique (the dive key), **#2 the Will** standing on the line outside him (the pitch key), **#3 the strong safety** in the alley (the lead halfback's block), and **#4 the cornerback** (the split end's block). Use the quarterback's two reads in order:

1. **#1 squeezes the dive**, so the quarterback pulls the ball. The fullback carries out his fake and takes #1 with him.
2. **#2 widens to take the pitch man**, so the quarterback keeps and turns up inside him.

The quarterback carries it, up the seam between #1 (who is tackling a fullback with no ball) and #2 (who is covering a halfback with no ball), with #3 and #4 blocked outside. A defense that does the same thing every snap has stopped playing option defense, which is why wishbone teams ran the triple until a defender changed his mind, and why defensive coaches, as 03-04 showed, learned to mix up who took whom.

Answer 3.

College: a legal 17-yard completion. The guard is inside the three-yard limit, and a pass that crosses the neutral zone is exactly what the college rule allows him to be up to three yards beyond. One foot in bounds is a catch in college.

NFL: no gain, and a flag. The guard is more than a yard downfield when a forward pass is thrown, so it is **illegal man downfield**: five yards from the previous spot, the completion erased, and the down replayed (04-06). And even without the flag, one foot in bounds isn't a catch in the NFL; the officials would rule it incomplete.

That's the whole reason NFL RPOs look different: the throw comes out in about a second, to a slant, a glance or a bubble, before any lineman can drift a yard, and the deep seam that a college team runs off the same action shows up on Sunday only as play-action, with the linemen pass-blocking.